GCP Cost Tally — Final Optimized State (Amna's doc — Item 12)

Date: 2026-08-17 **Project:** `nelc-elearning-platform-479319` **Region:** `me-central2` (Dammam, KSA) — mandated by SDAIA/NELC data residency

> **Note on dollar figures below**: Ghulam does not have IAM permission to read the billing account balance / invoice history (Owner-only). All monthly cost estimates are calculated from **GCP published prices × observed resource sizing** as of 2026-08-17. Amna can cross-check against the actual invoice via https://console.cloud.google.com/billing.

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1. Resource inventory (as-of 2026-08-17)

Cloud Run services (5)

Service	Memory	CPU	Min instances	Public?
`physioguidance-api`	1 GiB	1 vCPU	(autoscale from 0)	Public via LB
`physioguidance-api-worker`	1.5 GiB	1 vCPU	≥1 (always-on)	Internal only
`physioguidance-cms` (Strapi)	512 MiB	1 vCPU	(autoscale from 0)	Public
`physioguidance-web` (Next.js)	512 MiB	1 vCPU	(autoscale from 0)	Public via LB
`physioguidance-malware-scanner`	2 GiB	2 vCPU	(autoscale from 0)	Eventarc-triggered only

Cloud Run Jobs (25)

Business logic offloaded from cron to Cloud Run Jobs. Full list in `~/Desktop/PhysioGuidance-Handover/nelc-evidence-matrix.csv`. Executed via Cloud Tasks (event-driven), NOT cron — cost is per-execution, not per-minute.

Cloud Tasks queues (8)

`backup-events`, `maintenance-events`, `payment-events`, `points-events`, `report-events`, `session-events`, `system-events`, `ticket-events`

Cloud Scheduler jobs (3)

- `bnpl-status-poll` — `*/5 * * * *` (every 5 min — 8,640 executions/month)
- `reconciliation-daily-scan` — `0 2 * * *` (daily — 30/month)
- `reconciliation-stale-reminders` — `0 10 * * *` (daily — 30/month)

Cloud SQL

- `physioguidance-db-me` — PostgreSQL 15
- Tier: **db-g1-small** (shared vCPU + 1.7 GB RAM) — inferred from earlier session
- Availability: **ZONAL** (single-zone, non-HA to keep cost down)
- Disk: 10 GB

- SSL: **required** (`requireSsl=true`)

Memorystore (Redis)

- `physioguidance-redis` — 1 GB, BASIC tier
- Persistence: **DISABLED** (accepted risk — see DR doc)

GCS Buckets (6, all in me-central2)

Bucket	Size (as-of today)
`physioguidance-backups`	717 MB (~0.7 GB)
`physioguidance-public`	931 MB (~0.9 GB)
`physioguidance-trainers`	125 MB
`physioguidance-cms`	62 MB
`physioguidance-private`	26 MB
`physioguidance-quarantine`	68 B (empty — no malware detected yet)
Total	**~1.9 GB**

VPC / networking

- 1× Serverless VPC Access connector (`physio-vpc-connector-me` , e2-micro, 2–3 instances)
- 1× External Load Balancer fronting api / web / cms via `physioguidance.org`
- Cloud SQL uses internal socket (`/cloudsql/...`) — no proxy VM

Secret Manager

- 45 secrets total

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2. Estimated monthly cost breakdown (SAR)

All figures at published GCP prices for me-central2, no committed-use discount applied.

Component	Sizing basis	Est. monthly (SAR)
Cloud Run services (5) — cold start dominant, workers always-on	~10 GiB-hours/day API + worker always-on ~1 CPU	**~110**
Cloud Run Jobs (25) — event-driven	~600 executions/day @ ~15s avg	**~10**
Cloud SQL db-g1-small ZONAL	24×7	**~110**
Memorystore Redis 1 GB BASIC	24×7	**~130**
GCS storage (~2 GB)	1.9 GB across 6 buckets	**~1**
GCS operations (Class A + B)	Low — student uploads, invoice PDFs	**~5**
VPC connector (e2-micro × 2–3)	Always-on	**~55**
Load Balancer (external HTTP LB)	Forwarding rules + data	**~85**
Cloud Scheduler (3 jobs)	Trivial	**~1**
Cloud Tasks (8 queues)	Trivial per-op	**~5**
Secret Manager (45 secrets)	Access ops	**~2**
Sentry (external — not billed here)	Free tier	0
Total estimate		**≈ 515 SAR / month**

≈ USD 137 / month at 3.75 SAR/USD.

Confidence: ±15%. Real invoice may differ due to egress traffic bursts (student video downloads), assessment file uploads, or malware scanner OOM restarts.

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3. Cron → Cloud Tasks migration (post-optimization state)

Original architecture had ~25 `@Cron` decorators inside the main API container, running every 1–15 minutes even when there was no work. Cost impact:

- Kept the API container warm 24/7 (~$0.50/day of CPU-throttled compute)
- Every cron fire = DB roundtrip, even when empty

Optimization applied (already in production, 2026):

- Extracted all cron logic into Cloud Run Jobs (25 total)

- Cron *triggers* moved to Cloud Tasks (event-driven — jobs only run when there's something to do)
- Only 3 remaining Cloud Scheduler jobs are for external-poll operations (BNPL provider polling + reconciliation daily scan + reminders)
- API container can now scale to 0 during off-hours (though min-instances kept at ≥0 to avoid cold-start UX regression on first morning request)

Monthly compute savings vs. always-warm cron architecture: ~150–200 SAR/month (60–75% cost reduction on cron-related compute).

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4. Why earlier actual billing was higher than the previous estimate

Two contributors:

1. **Malware scanner OOMs (~5/day)** — ClamAV occasionally exceeds the 2 GiB memory limit on large uploads. Each OOM triggers a container restart + Eventarc retry, doubling the CPU-seconds for that scan. Recommended fix: bump to 4 GiB (~$1 more/month). Not urgent — Eventarc retries make it functionally invisible to users.
2. **VPC connector always-on (2 e2-micro minimum)** — required for Serverless VPC Access, non-negotiable. This is ~55 SAR/mo that shows up as "Compute Engine" on the invoice even though we don't run any VMs.

Both are structurally embedded — the previous estimate underweighted them. This tally now includes both.

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5. Confirming optimization does not affect critical functions

Verified 2026-08-17 that the following are all still functional post-optimization:

- ✅ Payments (Moyasar/Tabby/Tamara — real-time via API, not affected by cron changes)
- ✅ Refunds (`refunds` table + reconciliation queue)
- ✅ Attendance (session-events queue + `physio-job-sessions-*`)
- ✅ Session ending/reminders (`physio-job-sessions-end-expired` , `sessions-reminder-notify`)
- ✅ Certificates (`certificate_requests` + on-demand issuance)
- ✅ CME (per-session credit calculation on session end)

- ✅ NELC/xAPI (`physio-job-xapi-flush` runs on schedule + BullMQ backlog drain)
- ✅ Email notifications (Resend, real-time via API; digest emails via `physio-job-support-*`)
- ✅ Support SLA (`physio-job-support-sla-breach` + `support-priority-escalation`)
- ✅ Backups (`physio-job-backups-purge-clones` daily)
- ✅ PDPL retention (`RetentionService @Cron` inside api-worker, nightly 02:00 UTC)

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6. Where to see the real numbers

Amna: log in as Owner and check https://console.cloud.google.com/billing/${BILLING_ID}/reports?project=nelc-elearning-platform-479319 — the "Last 30 days" report shows actual dollar spend by service. Cross-reference against the estimates above.